

Complete Bounded Lozi Dynamics: Hyperbolic Coding and an Instability-Clock Obstruction

HCS-C385 theorem and reproducibility package

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Abstract

For the area-preserving Lozi family $F_a(x, y) = (1 - a|x| - y, x)$ with $a > 4$, we reconstruct the entire set of bounded two-sided orbits by a uniform contraction on sequence space. Every binary sequence occurs once, and the whole bounded set stays a positive distance from the nonsmooth switching line. This resolves global completeness, not merely a finite periodic prefix. Projective cone contraction gives the invariant stable and unstable lines. Every period, primitive cycle, signed reciprocal multiplier and reversal then follows with an exact affine-return formula. We distinguish the elementary unweighted zeta from a stability-weighted holomorphic germ with a controlled tail. The derivative also defines a separate positive instability roof. At integral parameters its periodic lengths cannot, after any positive rational rescaling, equal the logarithm of an integer greater than one. Exact rational and independent symbolic checks audit the normalization. These results establish source dynamics and an arithmetic obstruction, not a target spectral identity.

Keywords: Lozi map; bounded orbit; symbolic coding; uniform hyperbolicity; primitive cycle; instability clock.

中文摘要

本文对参数大于四的保面积 Lozi 映射，利用双向序列空间上的一致压缩，重建全部有界轨道，并证明每个二进制符号序列恰好对应一个状态。有界不变集与不可微切换线具有严格正间隔，因而全局完备性不依赖有限周期外推。进一步从射影锥压缩构造稳定与不稳定方向，得到所有周期、本原轨道、带符号互逆乘子和时间反演的精确分类。无权重 ζ 与稳定性加权解析芽分别定义。由导数产生的另一种不稳定性时钟，在整参数下不可能经正有理数缩放后精确等于大于二的整数的对数。这是一项源系统定理及算术限制，不构成目标零点或算子对应。

关键词：Lozi映射；有界轨道；符号编码；一致双曲性；本原周期；不稳定性时钟。

1 A global question beyond finite word enumeration

Finite sign words can be solved exactly without showing that every infinite itinerary exists or that every bounded geometric orbit has been found. For a nonsmooth map, a coding must also resolve derivatives at its branch boundary. We close these questions in one explicit sufficient chamber of the globally defined map

$$F_a(x, y) = (1 - a|x| - y, x), \quad a > 4. \quad (1)$$

Its inverse is $(x, y) \mapsto (y, 1 - a|y| - x)$. Let K_a be the set of points whose full two-sided orbit is bounded. The source clock is an application of (1). We do not identify that clock with any later suspension.

Lozi symbolic dynamics and kneading methods have established owners [1, 2]. Our contribution is a directly proved, normalized chamber theorem with independent exact evidence; no priority for the horseshoe mechanism or cone method is claimed. Earlier project C116 certified a pruned finite prefix at $(a, b) = (2, 1/2)$ of $(1 - a|x| + by, x)$. It did not prove the present statement. The constant-diagonal affine horseshoe in C127 also does not provide the word-dependent derivative matrices below.

2 All bounded two-sided orbits

Put $\Sigma = \{-1, +1\}^{\mathbb{Z}}$, with the left shift σ , and define

$$R = \frac{1}{a-2}, \quad q = \frac{2}{a}, \quad \delta = \frac{a-4}{a(a-2)} > 0.$$

For $s \in \Sigma$ let

$$(\mathcal{I}_s x)_j = \frac{s_j}{a} (1 - x_{j-1} - x_{j+1}) \quad \text{on } \ell^\infty(\mathbb{Z}).$$

Theorem 1. *There is a homeomorphism $\pi_a : \Sigma \rightarrow K_a$ satisfying $F_a \pi_a = \pi_a \sigma$. Every coordinate in the coded orbit obeys $\delta \leq s_j x_j \leq R$. These are all bounded two-sided orbits in the plane.*

Proof. The operator $\mathcal{J}_s$ contracts by $q < 1$ and preserves the closed radius- R ball because $(1 + 2R)/a = R$. Its unique fixed sequence satisfies

$$s_j x_j = \frac{1 - x_{j-1} - x_{j+1}}{a} \geq \frac{1 - 2R}{a} = \delta.$$

Hence its signs are exactly s_j and it solves $x_{j+1} = 1 - a|x_j| - x_{j-1}$. Conversely, a bounded orbit with $M = \sup_j |x_j|$ satisfies $aM \leq 1 + 2M$, so $M \leq R$. The same sign bound applies, and contraction uniqueness recovers that orbit from its sign sequence.

Set $\pi_a(s) = (x_0, x_{-1})$. Starting the contraction at zero, the N th approximation to either coordinate depends only on a finite symbol window and has error at most Rq^N . This proves continuity in the product topology. A state determines the full orbit under the continuous inverse of F_a , so distinct sign sequences cannot map to the same state. Exhaustion gives surjectivity. A continuous bijection from compact Σ to its Hausdorff image is a homeomorphism; the recurrence proves shift equivariance. $\square$

The coding is also uniform in parameters: comparing the fixed-point equations for $a, b > 4$ yields

$$\|x_a(s) - x_b(s)\|_\infty \leq \frac{|a - b|}{(a - 2)(b - 2)}. \quad (2)$$

Indeed, after moving the $2/a$ difference term to the left, the remaining bound is $|a^{-1} - b^{-1}|(1 + 2/(b - 2))$. The chamber boundary is deliberately not called sharp: at $a = 4$ this argument has zero sign margin and supplies no conclusion.

3 Variable-matrix hyperbolicity and complete primitive data

Define $r = (a - \sqrt{a^2 - 4})/2$, so $a - r = r^{-1}$ and $0 < r < 1$. On the smooth sign branch s , the derivative is

$$A_s = \begin{pmatrix} -as & -1 \\ 1 & 0 \end{pmatrix}, \quad \det A_s = 1.$$

Theorem 2. *The set K_a is uniformly hyperbolic. The derivative has continuous transverse invariant lines, with stable forward contraction and unstable backward contraction bounded by r^n in the max norm. For every $n \geq 1$, the whole plane has exactly 2^n points fixed by F_a^n .*

Proof. For $|v| \leq r|u|$, the derivative gives $|u'| \geq (a - r)|u| = r^{-1}|u|$ and $|v'| \leq r|u'|$. The slope map $t = v/u \mapsto (-as - t)^{-1}$ preserves $[-r, r]$ and contracts by at most r^2 . Composing from the infinite past gives a uniformly convergent continuous unstable slope. The inverse matrices preserve the cone $|u| \leq r|v|$; composing their slope maps from the infinite future gives the stable line with inverse-time expansion at least r^{-1} . The two cones meet only at zero. In each cone its dominant coordinate equals the max norm, giving the stated estimates with constant one. Theorem 1 keeps K_a away from $x = 0$, so these are actual derivatives, not derivatives assigned to singular points. The conjugacy gives 2^n periodic states. Every periodic orbit is bounded, so none has been missed outside K_a . $\square$

For a based word $w = s_0 \dots s_{n-1}$, compose in forward time:

$$M_w = A_{s_{n-1}} \dots A_{s_0}, \quad c_w = \sum_{j=0}^{n-1} A_{s_{n-1}} \dots A_{s_{j+1}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Empty products are identities. Hyperbolicity makes $I - M_w$ invertible, and the unique based state is $(I - M_w)^{-1} c_w$. Its least period is exactly the least repetition period of w . Therefore the number of oriented primitive cycles is

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}.$$

Let $\tau_w = \text{tr } M_w$. The two real multipliers are reciprocal, with

$$|\lambda_u(w)| \geq r^{-n}, \quad \text{sign } \lambda_u(w) = (-1)^n \prod_j s_j, \quad \lambda_s(w) = \lambda_u(w)^{-1}. \quad (3)$$

The sign follows from the sign of each dominant-coordinate factor $-as_j - t_j$. Repeating w raises M_w and its multipliers to powers. In particular the repeated traces satisfy $\tau_{k+1} = \tau_w \tau_k - \tau_{k-1}$, $\tau_0 = 2$, $\tau_1 = \tau_w$.

The involution $J(x, y) = (y, x)$ obeys $JF_a J = F_a^{-1}$. It reverses primitive words up to rotation and conjugates a return matrix to its inverse. We retain both orientations rather than dividing the primitive count by reversal. Branchwise determinant one and the global piecewise-affine bijection also give area preservation.

4 Separate source generating objects

The unweighted Artin–Mazur zeta is

$$Z_a(z) = \exp \left(\sum_{n \geq 1} \frac{2^n z^n}{n} \right) = \frac{1}{1 - 2z}. \quad (4)$$

Its ordinary primitive product converges for $|z| < 1/2$; the rational expression supplies its meromorphic continuation. It discards stability.

To retain the actual stability denominators, define instead

$$b_n = \sum_{x \in \text{Fix}(F_a^n)} \frac{1}{|\det(I - DF_a^n(x))|}, \quad d_a(z) = \exp \left(- \sum_{n \geq 1} \frac{b_n z^n}{n} \right).$$

These definitions do not assert a trace-class operator realization. If $L = |\lambda_u| \geq r^{-n}$, both possible signs of the multiplier satisfy

$$|\det(I - M_w)| \geq L(1 - L^{-1})^2, \quad 0 < b_n \leq \frac{(2r)^n}{(1 - r^n)^2}.$$

Thus d_a is holomorphic and nonzero for $2r|z| < 1$. For its logarithm, the omitted terms after order N have absolute value at most

$$\frac{(2r|z|)^{N+1}}{(N+1)(1-r)^2(1-2r|z|)}. \quad (5)$$

Absolute convergence permits primitive-orbit regrouping, but repetition k uses $|\det(I - M_\gamma^k)|^{-1}$, not an arbitrary scalar weight raised to the k th power. No continuation outside this disk is claimed.

5 An instability roof with no exact rationally scaled integer clock

Let $t_0^u(s)$ be the unstable slope. The positive continuous roof

$$\ell(s) = \log | -as_0 - t_0^u(s) | \geq \log r^{-1}$$

is derived from the source derivative. Its periodic sum is $\log |\lambda_u(w)|$, by multiplying the dominant-coordinate factors. It defines a new suspension, not the one-step time used in (4).

Theorem 3. *For integral $a \geq 5$, every positive rational c and every periodic word w satisfy*

$$c \log |\lambda_u(w)| \neq \log m \quad (m \in \mathbb{Z}, m > 1).$$

Proof. The matrix M_w belongs to $SL_2(\mathbb{Z})$ and has $|\tau_w| > 2$. For $T = |\tau_w| \geq 3$, the integer $T^2 - 4$ lies strictly between $(T-1)^2$ and T^2 , hence is nonsquare. The multiplier is a quadratic algebraic integer unit with nontrivial conjugate its reciprocal. If $|\lambda_u|^h$ were rational for $h \geq 1$, conjugation would equate it to $|\lambda_u|^{-h}$, contrary to expansion. Writing $c = h/j$, the displayed equality would give $|\lambda_u|^h = m^j$, a contradiction. $\square$

The result excludes primes and composites equally. It does not rule out approximate matching, irrational clock rescaling or a different roof. The nonintegral rational parameters in the numerical audit are not included in this unit theorem.

6 Independent evidence and integrity boundaries

The frozen rational grid is $a \in \{9/2, 5, 6\}$ with every word of length one through seven. The producer solves a two-dimensional affine return. A separate checker solves the full cyclic coordinate system by rational Gaussian elimination and independently evolves two tangent vectors.

Audit population	Count
Exact based returns	762
Primitive rows across parameters	123
Parameter–period summaries	21
Symbolic identities	19
Multiplier cells at 80 digits	762
Repaired-hash / serialization attacks	26 / 10

The multiplier normalization has maximum relative defect about 2.045×10^{-81} in the executed high-precision audit. All 36 hostile inputs were rejected. These finite computations audit formulas; the unbounded quantifiers come from the proofs. Byte replay, smoke tests and deterministic PDF builds are included in the release procedure. An internal author-swapped review checked the proof; it is not human or external cross-model peer review. Final review reproduced a boolean–integer coercion gap in the standalone checker. Explicit type checks and two repaired-hash regression attacks closed it; the mathematical evidence did not change.

The strict Route-A tuple, in A0–A4 order, is

(FAIL, WEAK, FAIL, FAIL, FORMAL_HINT).

Complete source orbits do not supply the missing arithmetic carrier or target divisor. The elementary source zeta is unchanged across parameters while the geometric multipliers vary, a direct control against mistaking symbolic counts for arithmetic validation.

Scope. NO_BAD_EULER_OR_ROOT_NUMBER. No target Euler factors, root numbers, automorphy, target divisor or functional equation, target-zero match or Hilbert–Pólya operator is claimed. Route B remains disabled.

7 Conclusion

The contraction theorem exhausts the global bounded set and resolves the switching-line issue in an explicit sufficient chamber. Variable signed stability, primitive repetition and reversal are then controlled for every period, rather than guessed from a prefix. The associated instability clock has an exact integral-unit limitation. Source generating functions with different weights or clocks remain separate; further arithmetic progress requires a new mechanism.

References

- [1] M. Misiurewicz and S. Štimac, *Symbolic dynamics for Lozi maps*, arXiv:1601.01745 (2016). Primary preprint.
- [2] W.-M. Zheng and J.-x. Liu, Symbolic analysis of attractor geometry for the Lozi map, *Physical Review E* **50**, 3241 (1994). doi:10.1103/PhysRevE.50.3241.

Round two: source germs, arithmetic obstruction and audit.